Calculus I Trigonometric Identities to Memorize

Larson 12th / Cengage-focused study sheet

1. Reciprocal Identities

$$\csc x = \frac{1}{\sin x} \qquad \sin x = \frac{1}{\csc x}$$

$$\sec x = \frac{1}{\cos x} \qquad \cos x = \frac{1}{\sec x}$$

$$\cot x = \frac{1}{\tan x} \qquad \tan x = \frac{1}{\cot x}$$

2. Quotient Identities

$$\tan x = \frac{\sin x}{\cos x} \qquad \cot x = \frac{\cos x}{\sin x}$$

3. Pythagorean Identities

$$\sin^2 x + \cos^2 x = 1$$

$$1 + \tan^2 x = \sec^2 x$$

$$1 + \cot^2 x = \csc^2 x$$

4. Useful Rearrangements

$$1 - \sin^2 x = \cos^2 x \qquad 1 - \cos^2 x = \sin^2 x$$

$$\sec^2 x - 1 = \tan^2 x \qquad \csc^2 x - 1 = \cot^2 x$$

$$1 - \sec^2 x = -\tan^2 x \qquad 1 - \csc^2 x = -\cot^2 x$$

5. Even / Odd Identities

$$\sin(-x) = -\sin x \qquad \cos(-x) = \cos x$$

$$\tan(-x) = -\tan x \qquad \csc(-x) = -\csc x$$

$$\sec(-x) = \sec x \qquad \cot(-x) = -\cot x$$

Memory rule: cos and sec are even; the other four trig functions are odd.

6. Double-Angle Identities

$$\sin(2x) = 2 \sin x \cos x$$

$$\cos(2x) = \cos^2 x - \sin^2 x$$

$$\cos(2x) = 2\cos^2 x - 1$$

$$\cos(2x) = 1 - 2\sin^2 x$$

$$\tan(2x) = \frac{2\tan x}{1 - \tan^2 x}$$

7. Power-Reduction / Half-Angle Forms

$$\sin^2 x = \frac{1 - \cos(2x)}{2}$$

$$\cos^2 x = \frac{1 + \cos(2x)}{2}$$

8. Sum and Difference Identities

$$\sin(a + b) = \sin a \cos b + \cos a \sin b$$

$$\sin(a - b) = \sin a \cos b - \cos a \sin b$$

$$\cos(a + b) = \cos a \cos b - \sin a \sin b$$

$$\cos(a - b) = \cos a \cos b + \sin a \sin b$$

$$\tan(a + b) = \frac{\tan a + \tan b}{1 - \tan a \tan b}$$

$$\tan(a - b) = \frac{\tan a - \tan b}{1 + \tan a \tan b}$$

9. Cofunction Identities

$$\sin\left(\frac{\pi}{2} - x\right) = \cos x$$

$$\cos\left(\frac{\pi}{2} - x\right) = \sin x$$

$$\tan\left(\frac{\pi}{2} - x\right) = \cot x$$

$$\sec\left(\frac{\pi}{2} - x\right) = \csc x$$

10. Core Seven - Highest Priority Review

$$\tan x = \frac{\sin x}{\cos x} \qquad \cot x = \frac{\cos x}{\sin x}$$

$$\sec x = \frac{1}{\cos x} \qquad \csc x = \frac{1}{\sin x}$$

$$\sin^2 x + \cos^2 x = 1$$

$$1 + \tan^2 x = \sec^2 x$$

$$1 + \cot^2 x = \csc^2 x$$

Also Memorize: Special Angles

x	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin x$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos x$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan x$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undef.

Unit-Circle Quadrant Signs

QI	all trig functions positive
QII	sine / cosecant positive
QIII	tangent / cotangent positive
QIV	cosine / secant positive

S | A
T | C

Fast Recognition for Calculus

- Convert tan, cot, sec, csc to sine/cosine when simplification stalls.
- Look first for $\sin^2 x + \cos^2 x$, $1 + \tan^2 x$, or $1 + \cot^2 x$.
- In implicit differentiation, reuse the *original equation* when it can replace a trig expression with x or y.

Study aid: this is a focused memorization sheet for common Calculus I work, not an official Cengage policy document.

Calculus I Derivative Rules to Memorize

Core differentiation formulas - combine with the chain rule when the input is a function

1. Basic / Linearity / Power Rules

$$\frac{d}{dx}[c] = 0$$

$$\frac{d}{dx}[x] = 1$$

$$\frac{d}{dx}[c\,f(x)] = c\,f'(x)$$

$$\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)$$

$$\frac{d}{dx}[f(x) - g(x)] = f'(x) - g'(x)$$

$$\frac{d}{dx}[x^n] = nx^{n-1}$$

$$\frac{d}{dx}[u^n] = nu^{n-1}u'$$

2. Product / Quotient / Chain

$$\frac{d}{dx}[uv] = u'v + uv'$$

$$\frac{d}{dx}\left[\frac{u}{v}\right] = \frac{vu' - uv'}{v^2}$$

$$\frac{d}{dx}[f(g(x))] = f'(g(x))\,g'(x)$$

3. Reciprocal / Root Special Cases

$$\frac{d}{dx}\left[\frac{1}{x}\right] = -\frac{1}{x^2}$$

$$\frac{d}{dx}[\sqrt{x}] = \frac{1}{2\sqrt{x}}$$

Notation: c is constant; u=u(x), v=v(x).

4. Exponential / Logarithm Rules

$$\frac{d}{dx}[e^x] = e^x$$

$$\frac{d}{dx}[a^x] = a^x \ln a$$

$$\frac{d}{dx}[\ln x] = \frac{1}{x}$$

$$\frac{d}{dx}[\log_a x] = \frac{1}{x \ln a}$$

$$\frac{d}{dx}[e^u] = e^u u'$$

$$\frac{d}{dx}[a^u] = a^u \ln(a)\,u'$$

$$\frac{d}{dx}[\ln|u|] = \frac{u'}{u}$$

$$\frac{d}{dx}[\log_a u] = \frac{u'}{u \ln a}$$

5. Trigonometric Derivatives

$$\frac{d}{dx}[\sin x] = \cos x$$

$$\frac{d}{dx}[\cos x] = -\sin x$$

$$\frac{d}{dx}[\tan x] = \sec^2 x$$

$$\frac{d}{dx}[\cot x] = -\csc^2 x$$

$$\frac{d}{dx}[\sec x] = \sec x \tan x$$

$$\frac{d}{dx}[\csc x] = -\csc x \cot x$$

Trig derivative formulas assume radian measure.

6. Inverse Trigonometric Derivatives

$$\frac{d}{dx}[\arcsin x] = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}[\arccos x] = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}[\arctan x] = \frac{1}{1+x^2}$$

$$\frac{d}{dx}[\operatorname{arccot} x] = -\frac{1}{1+x^2}$$

$$\frac{d}{dx}[\operatorname{arcsec} x] = \frac{1}{|x|\sqrt{x^2-1}}$$

$$\frac{d}{dx}[\operatorname{arccsc} x] = -\frac{1}{|x|\sqrt{x^2-1}}$$

7. Other Important Calculus I Rules

$$\frac{d}{dx}[f^{-1}(x)] = \frac{1}{f'(f^{-1}(x))}$$

$$\frac{d}{dx}[y] = y'$$

$$\frac{d}{dx}[u^v] = u^v\left(v' \ln u + v\frac{u'}{u}\right)$$

Fast Recognition

- If x is replaced by u(x), multiply by u'(x).
- Product: first' second + first second'.
- Quotient: bottom*top' - top*bottom', over bottom^2.
- Implicit differentiation: every derivative of y brings a y'.